Predicting Stellar Labels for the Sagittarius Stream Using Gaia DR3 XP Spectra

Giovanny Rosales, Hunter Morgan

November 2026

Abstract

High-resolution spectroscopic surveys provide precise stellar parameters and chemical abundances, but cover only a fraction of the stars observed by Gaia. This project transfers five stellar labels from APOGEE DR17 to Gaia DR3 BP/RP XP spectra and applies the selected model to candidate members of the Sagittarius stellar stream. The predicted labels are effective temperature, surface gravity, metallicity, nitrogen-to-oxygen abundance, and aluminum-to-iron abundance.

The completed workflow constructs a quality-filtered APOGEE–Gaia training catalogue, downloads available XP continuous coefficients in restart-safe batches, converts each BP/RP coefficient pair into 108 normalized features, creates fixed training, validation, and test partitions, and compares regularized linear regression, a randomized tree ensemble, and feedforward neural networks. Of 48,650 targeted XP sources in the completed download run, 44,650 were retrieved and passed preprocessing. These sources were divided into 31,255 training, 6,697 validation, and 6,698 test stars. The selected two-hidden-layer multilayer perceptron achieved test mean absolute errors of 76.32 K in $T_{\rm eff}$, 0.172 in $\log g$, 0.0838 dex in [Fe/H], 0.0978 dex in [N/O], and 0.0552 dex in [Al/Fe]. The model was then applied to 29,886 Sagittarius XP sources. Of these, 27,106 joined to the final Sagittarius crossmatch catalogue, and 26,787 remained after predictions outside the training-label ranges were flagged. The results demonstrate that nonlinear models substantially outperform the linear baseline for transferring APOGEE labels to Gaia XP representations.

1 Introduction

The chemical and dynamical properties of stars preserve information about the assembly of the Milky Way and its accreted satellites. Metallicity and elemental-abundance ratios can distinguish stellar populations, constrain enrichment histories, and identify debris associated with disrupted dwarf galaxies. The Sagittarius dwarf spheroidal galaxy and its tidal stream provide an important setting for studying chemical evolution during an ongoing accretion event.

APOGEE DR17 supplies near-infrared stellar parameters and abundances for a large spectroscopic sample [1]. Gaia DR3 provides BP/RP low-resolution spectrophotometry, commonly called XP spectra, for a much larger population [3, 2]. Previous work demonstrates that data-driven methods can infer stellar parameters and selected abundances from Gaia XP data [7, 8, 5].

The project goals are to:

1. construct an APOGEE DR17 and Gaia DR3 XP training sample;
2. predict $T_{\rm eff}$, $\log g$, [Fe/H], [N/O], and [Al/Fe] from normalized XP coefficients;
3. compare linear, tree-ensemble, and neural-network regression methods on fixed held-out data; and

4. apply the selected model to Sagittarius stream candidates with XP spectra while flagging extrapolations beyond the training-label domain.

2 Problem Definition

For each star i, the input is a vector of normalized Gaia XP coefficients, $\mathbf{x}_i \in \mathbb{R}^{108}$. The desired output is the five-dimensional label vector

$$\mathbf{y}_i = \left(T_{\rm eff}, \log g, [\mathrm{Fe/H}], [\mathrm{N/O}], [\mathrm{Al/Fe}]\right)_i . \tag{1}$$

The task is supervised multi-output regression. All candidate models receive identical input features, target labels, and source-level data partitions. Model selection uses validation performance, and the test partition is reserved for final held-out evaluation.

3 Data

3.1 Gaia DR3 XP Spectra

Gaia DR3 publishes BP and RP mean spectra using coefficients that multiply internally calibrated basis functions [2]. Each source supplies 55 BP and 55 RP coefficients. The first coefficient in each band is used to normalize the remaining coefficients:

$$\widetilde{c}_j^{\rm BP} = \frac{c_j^{\rm BP}}{c_0^{\rm BP}}, \qquad \widetilde{c}_j^{\rm RP} = \frac{c_j^{\rm RP}}{c_0^{\rm RP}}, \qquad j = 1, \ldots, 54. \tag{2}$$

The unit-valued first coefficient is omitted from each band, leaving

$$N_{\rm XP} = (55 - 1) + (55 - 1) = 108 \tag{3}$$

features per source. A row is rejected if either coefficient array has the wrong length, contains a nonfinite value, or has a zero first coefficient.

3.2 APOGEE DR17 Labels

The labels come from the APOGEE DR17 `allStarLite-dr17-synspec_rev1.fits` catalogue [1]. The workflow uses `TEFF`, `LOGG`, `FE_H`, `N_FE`, `O_FE`, and `AL_FE`. Nitrogen-to-oxygen abundance is derived as

$$[\mathrm{N/O}] = [\mathrm{N/Fe}] - [\mathrm{O/Fe}]. \tag{4}$$

The quality policy rejects `STAR_BAD`, requires `SNR`≥ 70, requires finite target labels, and initially restricts the catalogue to $3500 \leq T_{\rm eff} \leq 7000$ K and $0 \leq \log g \leq 5$. It also removes sources with target-related `TEFF_WARN`, `LOGG_WARN`, `M_H_WARN`, or `N_M_WARN` flags. The `VMICRO_WARN` flag is recorded but not used as an exclusion because microturbulence is not a prediction target.

3.3 Sagittarius Application Sample

The application sample originates from the Sagittarius catalogue of Vasiliev et al. [11]. The earlier pipeline linked Gaia DR2 identifiers to unique Gaia DR3 sources, retained candidates with XP continuous products, and downloaded the available XP coefficients. This application sample is not used to fit the models. Instead, it is processed with the feature transformation and scalers learned from the APOGEE training sample.

4 Training-Sample Construction

The training workflow began with 733,901 APOGEE catalogue rows. Table 1 summarizes the quality and crossmatch stages completed before coefficient retrieval.

Table 1: APOGEE DR17–Gaia DR3 XP training-metadata cutflow.

Selection	Before	After	Removed
Reject `ASPCAPFLAG STAR_BAD`	733,901	691,591	42,310
Require `SNR`≥ 70	691,591	597,478	94,113
Require five finite targets	597,478	495,611	101,867
Require $3500 \leq T_{\rm eff} \leq 7000$ K	495,611	495,611	0
Require $0 \leq \log g \leq 5$	495,611	495,464	147
Require valid unique Gaia source ID	495,464	448,728	46,736
Verify identifier in Gaia DR3	448,728	448,728	0
Require XP continuous availability	448,728	405,160	43,568
Reject target-related ASPCAP warnings	405,160	334,486	70,674

Because downloading all 334,486 XP products was not practical within the project schedule and Gaia DataLink availability was intermittent, the production download used a stratified 100,000-source design divided into core and extension selections. The completed run targeted 48,650 source identifiers in 973 batches of 50. A total of 893 batch files were successfully written, containing 44,650 unique sources. Eighty batches exhausted their retries, corresponding to 4,000 unavailable sources. The successful data contained zero duplicate identifiers, and every retrieved row contained valid 55-element BP and RP arrays. Therefore, all 44,650 retrieved sources passed coefficient preprocessing.

5 Preprocessing and Fixed Splits

The downloaded XP table was joined to the corresponding APOGEE metadata by Gaia `source_id`. Each valid 55+55 coefficient pair was transformed into 108 normalized features. The five targets and 108 inputs were checked for finite values, and source identifiers were preserved throughout preprocessing.

The 44,650 model-ready sources were divided with random seed 42 into non-overlapping 70%/15%/15% partitions. Table 2 gives the exact counts.

Table 2: Fixed source-level model partitions.

Partition	Sources
Training	31,255
Validation	6,697
Test	6,698
Total	44,650

For Ridge and the neural networks, a `StandardScaler` was fitted to the training features only. The five targets were also standardized using training-set statistics so the much larger numerical scale of $T_{\rm eff}$ did not dominate the multi-output objective. The validation and test partitions were

transformed with the fitted training scalers. Extra Trees used the unscaled 108 features because tree splits are invariant to monotonic rescaling. Split identifiers, fitted scalers, and the selected model checkpoint were saved for reproducibility.

6 Models and Experimental Setup

Four model configurations were evaluated. Three represent distinct algorithm families, while the fourth tests whether a deeper neural architecture improves performance.

6.1 Ridge Regression

Ridge regression provides a regularized linear baseline. The model used $\alpha = 1.0$ and predicted all five standardized targets jointly. This model tests whether a linear mapping from normalized XP coefficients is sufficient.

6.2 Extra Trees

The Extra Trees regressor [10] provides a nonlinear, non-neural comparison. The implementation used 300 trees, the square root of the 108 input features as the candidate feature count at each split, a minimum of two samples per leaf, random seed 42, and all available CPU cores.

6.3 Baseline MLP

The baseline neural network was implemented in PyTorch [9]. It contains 108 inputs, two 128-unit hidden layers, ReLU activations, dropout probability 0.2 after each hidden layer, and five linear outputs. Training used Adam with learning rate 10^{-3}, weight decay 10^{-5}, batch size 256, and mean-squared error on standardized targets.

6.4 Deeper MLP

The deeper architecture contains hidden layers of 256, 128, and 64 units, with the same activation, dropout, optimizer, and output structure. This model tests whether increased depth and capacity improve prediction accuracy.

Both neural networks used a maximum of 200 epochs and validation-based early stopping with patience 25. Checkpoints were scheduled at epochs 10, 25, 50, 100, and 200 when reached, while the lowest-validation-loss state was always preserved as the best model. Training used the Apple Metal Performance Shaders device. The baseline MLP stopped after 138 epochs and selected epoch 113. The deeper MLP stopped after 92 epochs and selected epoch 67.

7 Held-Out Model Results

Table 3 reports the final test MAE, RMSE, and coefficient of determination for all four configurations. Lower MAE and RMSE are better, while R^2 values closer to one indicate that more of the target variance is explained.

Table 3: Held-out test performance.

Model	Target	MAE	RMSE	R^2
Ridge	$T_{\rm eff}$	116.626	165.368	0.9185
	$\log g$	0.3465	0.4957	0.7682
	[Fe/H]	0.1559	0.2327	0.5480
	[N/O]	0.1326	0.1954	0.2223
	[Al/Fe]	0.0818	0.1221	0.2167
Extra Trees	$T_{\rm eff}$	96.996	143.015	0.9390
	$\log g$	0.2577	0.3619	0.8764
	[Fe/H]	0.1373	0.2028	0.6566
	[N/O]	0.1224	0.1832	0.3167
	[Al/Fe]	0.0715	0.1067	0.4022
Baseline MLP	$T_{\rm eff}$	**76.323**	**115.708**	**0.9601**
	$\log g$	**0.1721**	0.2518	0.9402
	[Fe/H]	0.0838	0.1281	0.8629
	[N/O]	**0.0978**	**0.1600**	**0.4787**
	[Al/Fe]	0.0552	0.0837	0.6324
Deeper MLP	$T_{\rm eff}$	76.630	115.837	0.9600
	$\log g$	0.1741	**0.2515**	**0.9403**
	[Fe/H]	**0.0830**	**0.1256**	**0.8683**
	[N/O]	0.0979	0.1623	0.4638
	[Al/Fe]	**0.0550**	**0.0822**	**0.6449**

Extra Trees improved upon Ridge for every target, demonstrating that the mapping from XP coefficients to stellar labels is not adequately described by the linear baseline. Both neural networks then substantially improved upon Extra Trees. The strongest gains occurred for $\log g$ and [Fe/H]. The abundance targets [N/O] and [Al/Fe] had lower R^2 values than temperature, gravity, and metallicity, indicating that these labels remain more difficult to infer from the available XP representation.

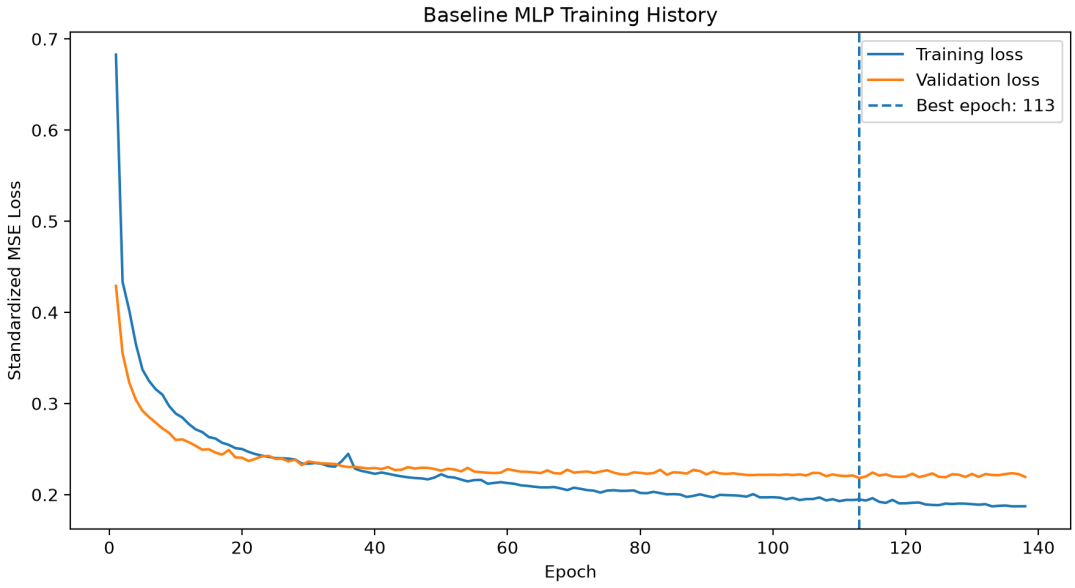
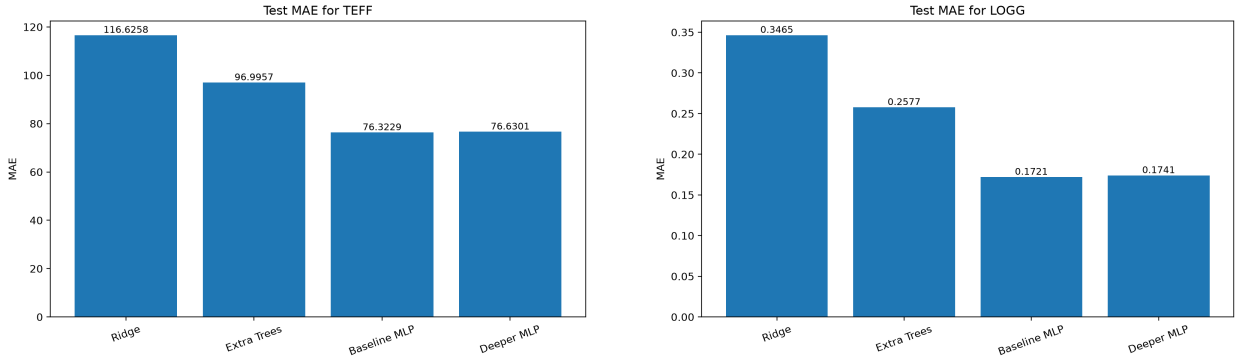
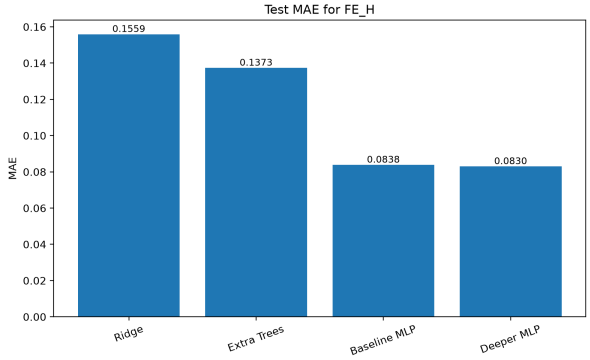


Figure 1: Training and validation loss for the baseline MLP. The selected checkpoint occurred at epoch 113, where validation loss was minimized before early stopping at epoch 138.

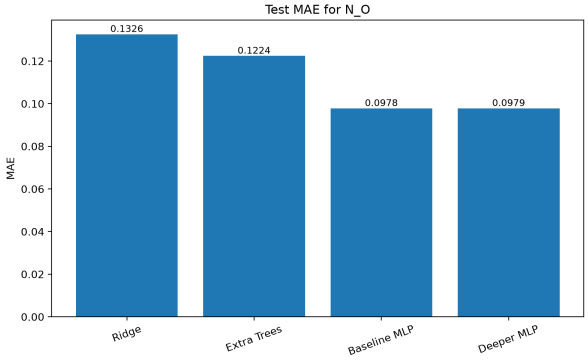


(a) T_{eff}

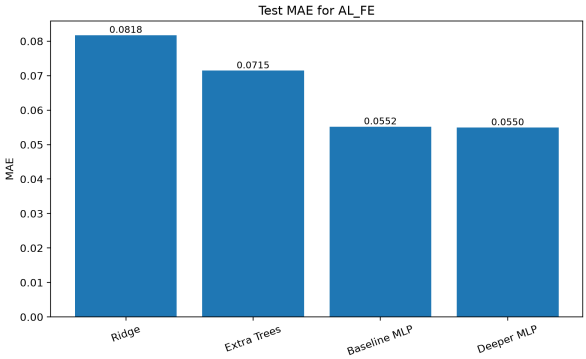
(b) $\log g$



(c) [Fe/H]



(d) [N/O]



(e) [Al/Fe]

Figure 2: Test-set MAE by model and target. The neural networks outperform Ridge and Extra Trees for all five labels.

8 Model Selection

The baseline MLP was selected using overall validation performance, not the test set. Its best standardized validation loss was 0.218281 at epoch 113, compared with 0.219965 for the deeper MLP at epoch 67. The deeper model produced marginally better held-out results for some individual quantities, especially [Fe/H] and [Al/Fe], but did not provide a clear overall improvement. The baseline MLP was therefore preferred because it achieved the lowest overall validation loss with a

smaller architecture.

The selected model trained in 31.34 seconds on the local Apple MPS device, averaging approximately 0.23 seconds per completed epoch. The short runtime is expected because the training matrix contains only 31,255 rows and 108 numeric features, and the network is relatively small. Data acquisition and catalogue construction were substantially more expensive than model fitting.

9 Sagittarius Inference

The selected baseline MLP, training-fitted feature scaler, and training-fitted target scaler were applied to the combined Sagittarius XP data. All 29,886 input rows passed coefficient preprocessing and received predictions for the five targets. The predictions were then joined to the final unique Sagittarius DR3 XP crossmatch catalogue using Gaia DR3 `source_id`.

Table 4 summarizes the application-stage accounting. The 2,780 predictions not found in the final 27,106-row crossmatch catalogue were retained in a separate audit file rather than included in the science catalogue. Within the matched sample, 319 rows had at least one predicted target outside the corresponding range observed in the 44,650-source training sample. These rows were flagged as extrapolations. The final conservative catalogue contains 26,787 matched sources whose five predictions all lie within the training-label ranges.

Table 4: Sagittarius prediction and catalogue accounting.

Stage	Sources
XP rows receiving predictions	29,886
Matched to final Sagittarius DR3 XP catalogue	27,106
Predictions unmatched to final catalogue	2,780
Matched rows outside at least one training range	319
Final in-domain Sagittarius science catalogue	**26,787**

Across all 29,886 raw predictions, the median values were approximately 4493 K in $T_{\rm eff}$, 1.74 in $\log g$, -0.56 dex in [Fe/H], 0.054 dex in [N/O], and -0.100 dex in [Al/Fe]. Most predictions were within the training domain, but extreme outputs included implausibly low and high temperatures and gravities. Range flagging was therefore necessary before scientific interpretation. In total, 355 of the 29,886 raw predictions, or 1.19%, fell outside at least one training-label range. After restricting to the matched Sagittarius catalogue, 319 such rows remained.

10 Limitations

The production model used 44,650 XP sources rather than the full 334,486-source quality-filtered metadata catalogue. This reduction resulted from the project download design, intermittent Gaia DataLink failures, and the decision to proceed with the successfully retrieved sources. The stratified selection improves coverage relative to an unstructured subset, but the model may still underrepresent rare regions of label space.

The deterministic models do not provide calibrated predictive uncertainties. Range checks identify obvious extrapolations, but a prediction inside the training-label minimum and maximum can still occur in a sparsely sampled part of the joint feature and label distribution. Future work could use ensemble variance, Monte Carlo dropout [4], or heteroscedastic regression [6] to estimate

uncertainty. The lower R^2 values for [N/O] and [Al/Fe] also indicate that abundance inference remains less constrained than temperature, gravity, and metallicity.

Finally, the 2,780 XP predictions not present in the final Sagittarius crossmatch catalogue were excluded from the science-ready product. They remain available as an audit sample, but should not be interpreted as confirmed members of the final Sagittarius selection without resolving their catalogue provenance.

11 Reproducibility

The repository contains runnable scripts for combining XP products, preprocessing coefficient arrays, creating deterministic source-level splits, training all model configurations, evaluating predictions, and applying the selected model to Sagittarius sources. The baseline MLP checkpoint and fitted feature and target scalers are saved with the result metadata. Eleven preprocessing tests pass with `pytest`.

The principal commands are:

```
python src/combine_training_xp.py [arguments]
python src/create_splits.py --input <metadata.csv> --seed 42
python src/train_model.py --model ridge
python src/train_model.py --model extra_trees
python src/train_model.py --model baseline_mlp --device mps
python src/train_model.py --model deeper_mlp --device mps
python src/predict_sagittarius.py [model and scaler arguments]
python -m pytest -q
```

Exact project-specific commands, required files, and expected outputs are documented in `README.md`. Large regenerable FITS batches and the model-ready NPZ are excluded from normal Git tracking, while compact split identifiers, metrics, figures, the selected checkpoint, scalers, and final Sagittarius catalogues are retained.

12 Conclusion

This project completed an end-to-end workflow for transferring APOGEE DR17 stellar labels to Gaia DR3 XP spectra and applying the selected model to Sagittarius candidates. A total of 44,650 successfully downloaded sources were converted into 108 normalized XP features and divided into fixed training, validation, and test partitions. Ridge, Extra Trees, and neural-network regressors were compared using identical data. Extra Trees consistently outperformed the linear baseline, while both MLPs achieved the strongest held-out results. The smaller baseline MLP was selected because it achieved the best overall validation loss without the additional complexity of the deeper network.

The selected model reached test R^2 values of 0.960 for $T_{\rm eff}$, 0.940 for $\log g$, 0.863 for [Fe/H], 0.479 for [N/O], and 0.632 for [Al/Fe]. It generated predictions for 29,886 Sagittarius XP sources. After joining to the final crossmatch catalogue and excluding range-flagged extrapolations, the resulting science-ready catalogue contains 26,787 sources. The completed code, model artifacts, metrics, figures, and catalogue products establish a reproducible basis for subsequent chemical analysis of the Sagittarius system.

Code and Data Availability

The implementation and reproduction instructions are available at https://github.com/LightsRedemption/DeepLearning_TeamAstro. Large downloaded and generated FITS products are retained locally and excluded from normal Git tracking because they can be regenerated from the documented pipeline.

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